EART11300

Physical properties of minerals

Video 2: Crystal habit

Reading: Section on habit in the online mineralogy textbook:
https://opengeology.org/Mineralogy/3-properties/#332_Mineral_Habit

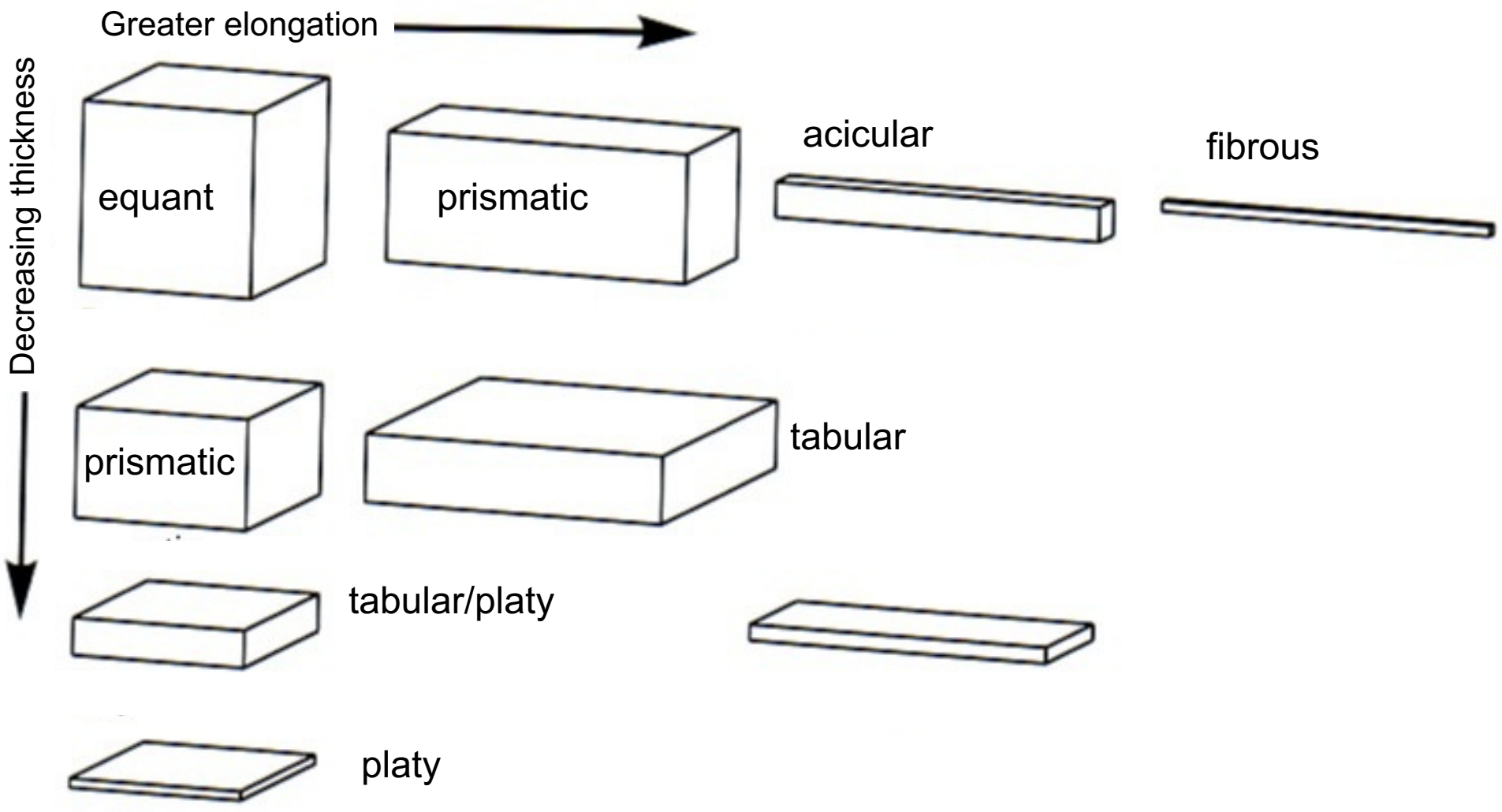
What is ‘habit’?

- The combination and relative development of crystal faces is called the crystal’s ‘habit’.
- Habit depends partly on crystal structure, but also on the way the crystal has grown, allowing some faces to develop preferentially.

Terms used to describe habit:

- equant
- tabular
- platy
- prismatic (moderately elongate)
- acicular (needle-shaped)
- fibrous
- Massive = compact mass of interlocking crystals with no clear division between individual grains.

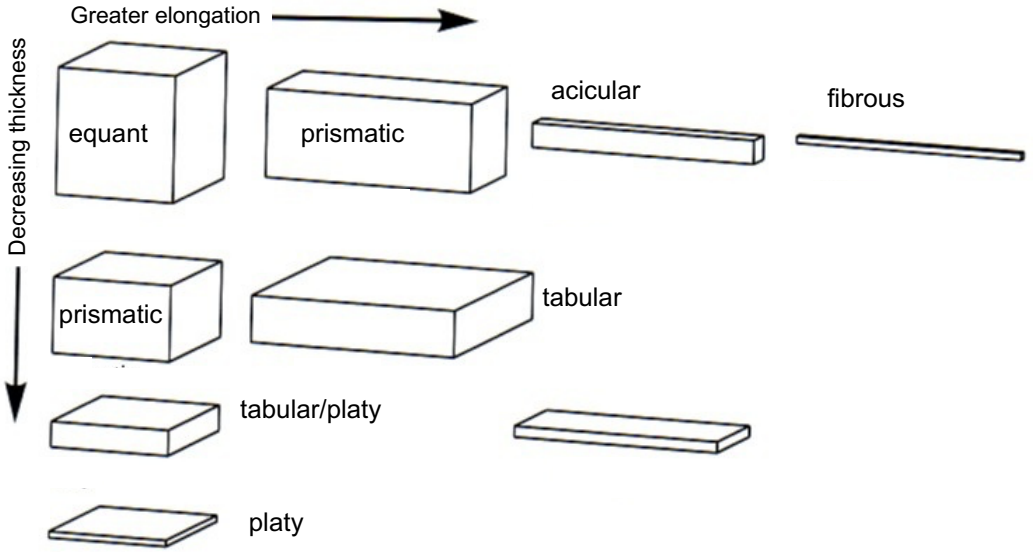
Crystal habits



Examples: 1. Fluorite CaF_2

- The ideal shape of a fluorite crystal is a cube.
- The habit that this corresponds to is?

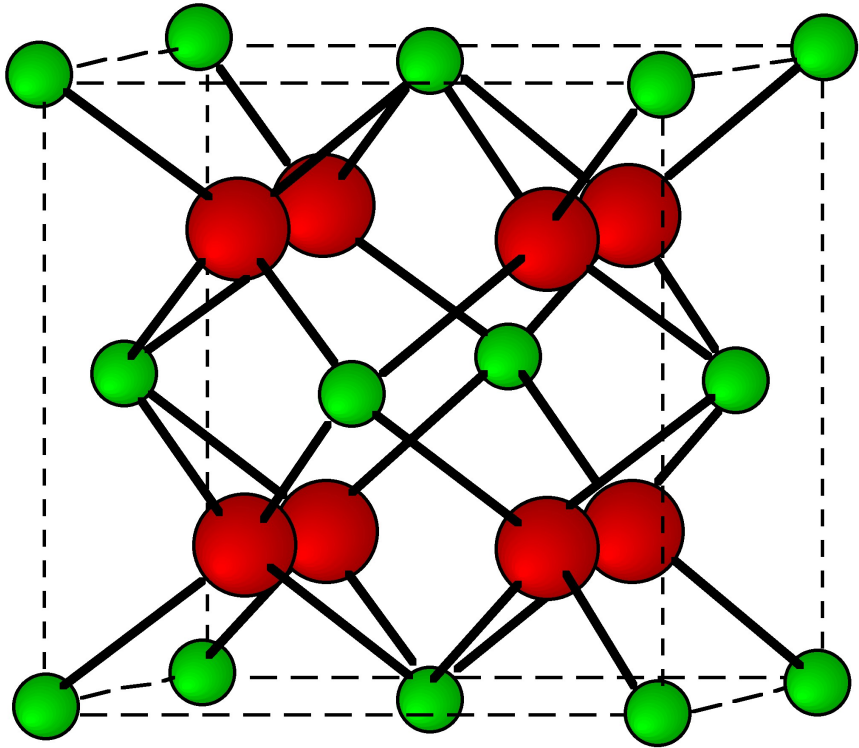




Examples: 1. Fluorite CaF_2

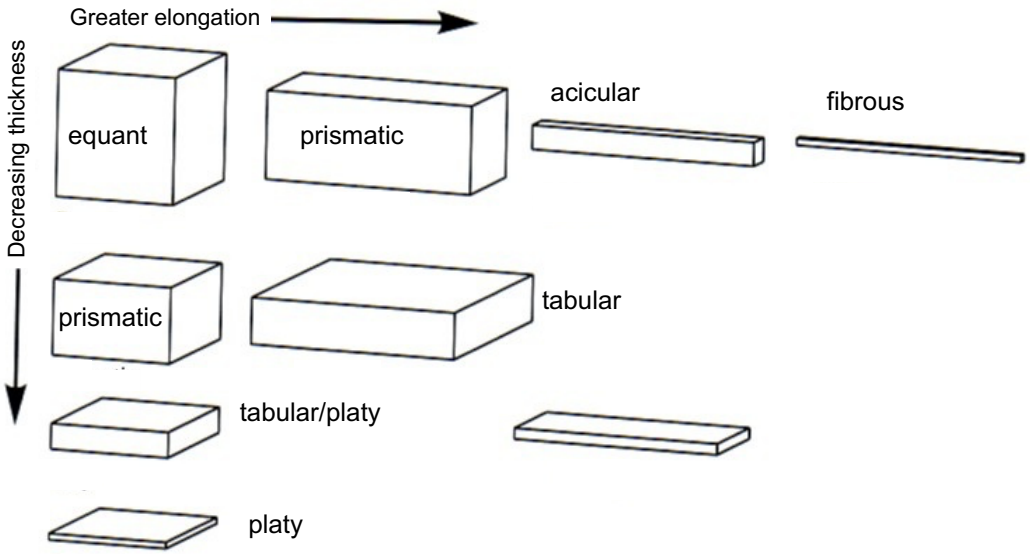
- The ideal shape of a fluorite crystal is a cube.
- The habit that this corresponds to is? **equant**
- Fluorite's habit is related to its crystal structure: the cube faces contain a high density of atoms. It is typically the case that the most prominent faces of a crystal are those with the highest density of atoms.





Examples: 2. Mica

- Mica is a common mineral in granites and metamorphosed mudstones.
- Its habit is?





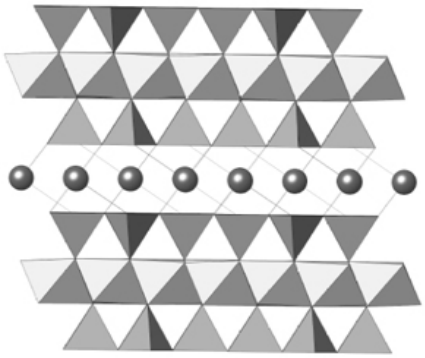
Examples: 2. Mica



- Mica is a common mineral in granites and metamorphosed mudstones.
- Its habit is **platy** (occasionally **tabular**).
- The mica structure is made of layers in which the bonds are short and strong (2 sheets of Si-O tetrahedra sandwiching a sheet of Mg or Al), separated by layers of large potassium ions.
- Because the potassium is weakly bonded, mica grows slowly in this direction, giving rise to very thin crystals.
- But it grows quickly parallel to the layers, because the atoms are denser in this plane, which means that bonds form faster allowing large platy crystals to form.

Looking down on a layer: strong bonds = rapid growth in all directions within the layer.

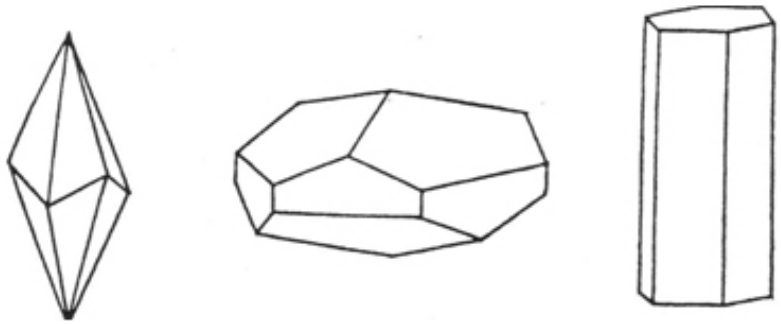


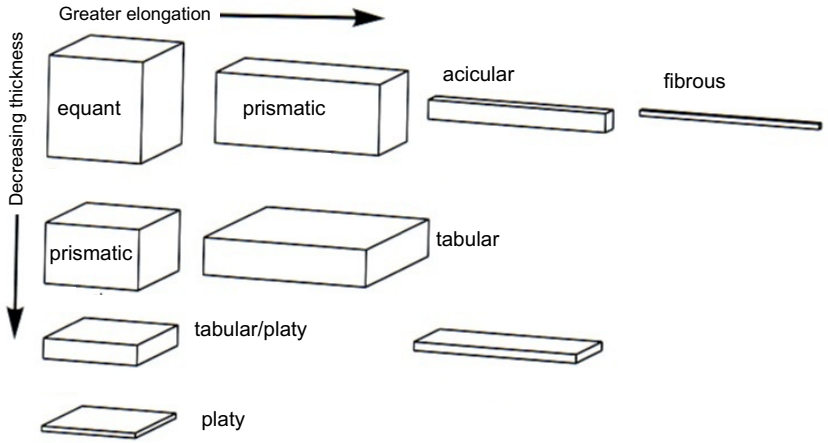


Looking at a layer side on: weak bonds to potassium = slow growth perpendicular to the layers.

Examples: 3. Calcite

- Calcite is found in sedimentary rocks – limestone – and metamorphosed limestone – marble.
- It crystallises with different habits under different conditions, e.g. see the figures below – you could use fancy terms for these habits, or simply say?



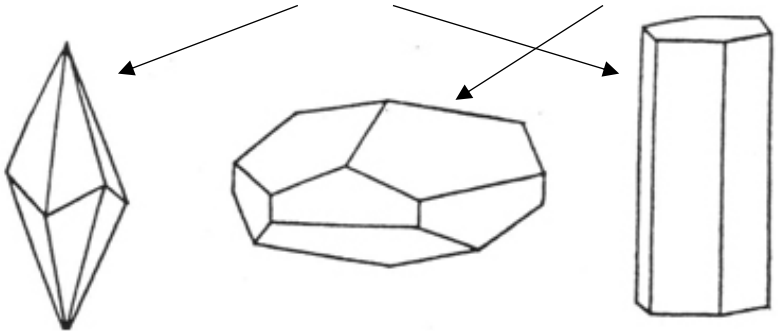






Examples: 3. Calcite

- Calcite is found in sedimentary rocks – limestone – and metamorphosed limestone – marble.
- It crystallises with different habits under different conditions, e.g. see the figures below – you could use fancy terms for these habits, or simply say **prismatic** and **tabular**.

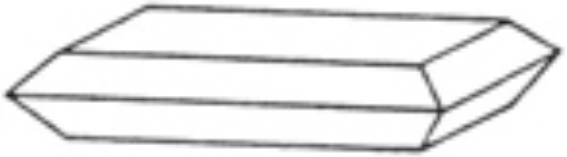


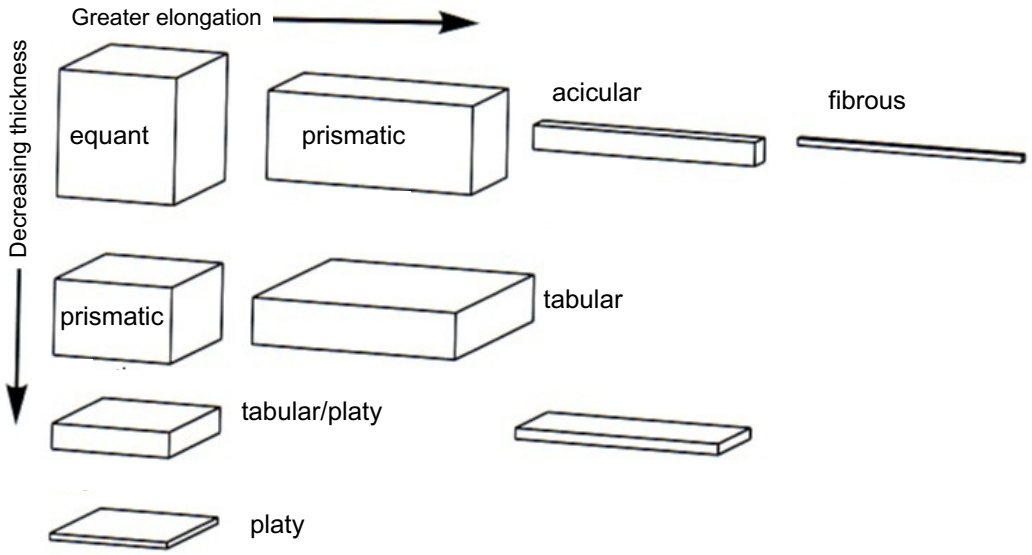




Examples: 4. Barite, $BaSO_4$

- Barite is common in veins in limestones.
- It also has a variable habit, but a common one is shown below. It is?

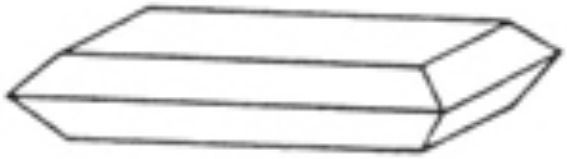


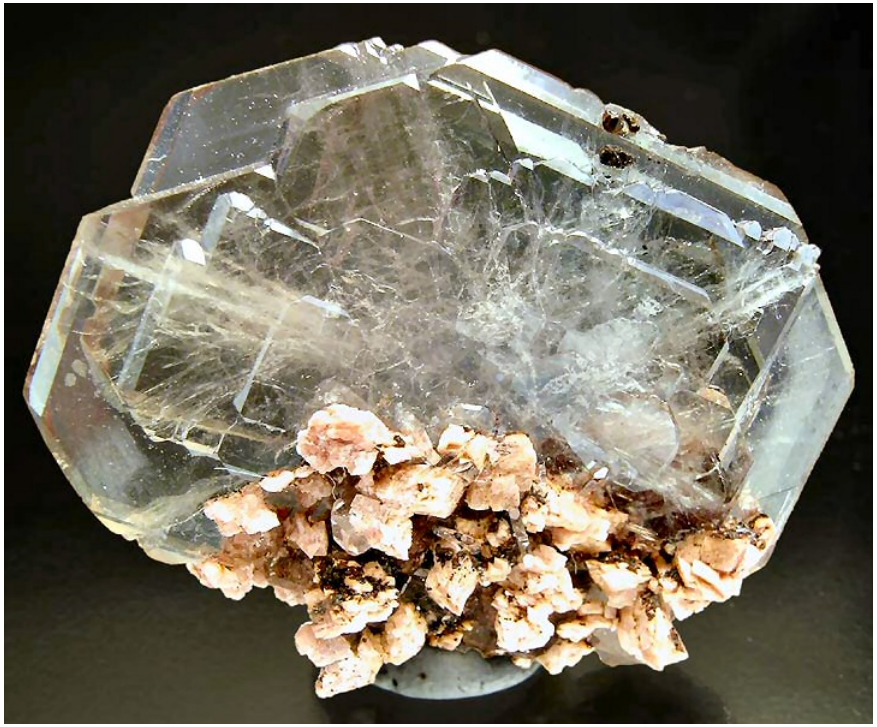




Examples: 4. Barite, $BaSO_4$

- Barite is common in veins in limestones.
- It also has a variable habit, but a common one is **tabular**:

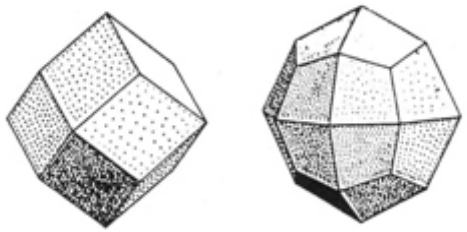




Examples: 5. Garnet, $(Ca,Mg,Fe)_3Al_2Si_3O_{12}$

- Garnet is common in high-grade metamorphic rocks.
- It grows with different shapes, due to different relative development of different faces (see examples on display and the diagrams below), but the habit is always the same: ?



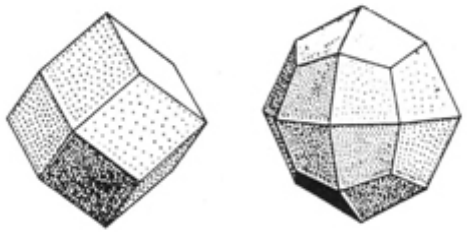




Examples: 5. Garnet, $(Ca,Mg,Fe)_3Al_2Si_3O_{12}$

- Garnet is common in high-grade metamorphic rocks.
- It grows with different shapes, due to different relative development of different faces (see examples on display and the diagrams below), but the habit is always the same: **equant**.

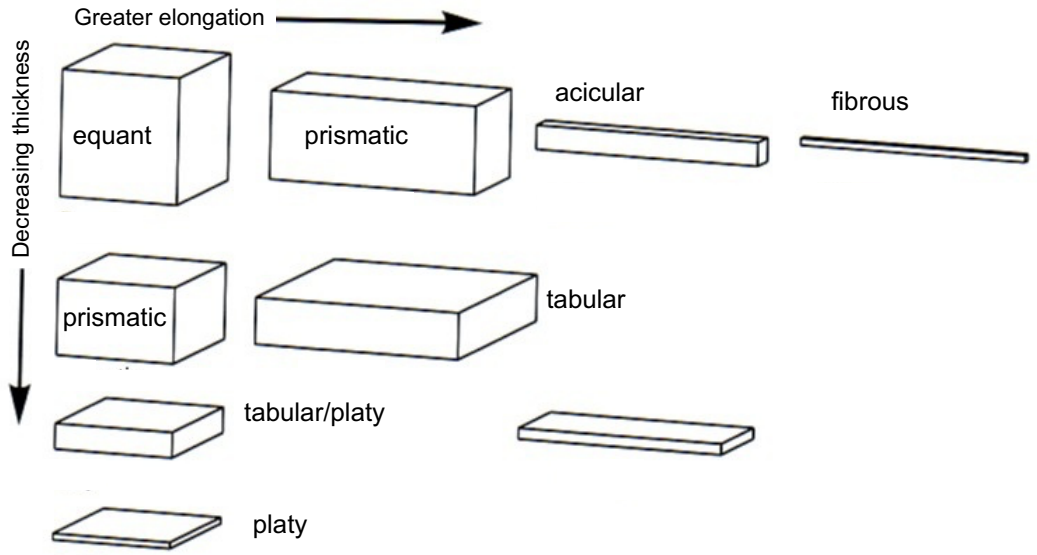






Examples: 6. Quartz, SiO_2

- Quartz grows with a characteristic habit when it is allowed to grow freely in an open space (e.g. in a fracture in a rock through which hot fluids are circulating).
- This habit is ?





Examples: 6. Quartz, SiO_2

- Quartz grows with a characteristic habit when it is allowed to grow freely in an open space (e.g. in a fracture in a rock through which hot fluids are circulating).
- The habit is **prismatic**

